

HDOC Day-12

Question: Menu-Driven Program for Digit Analysis Using Loops & Switch-Case

1. Problem Statement & Identification

- **Problem:** Create a menu-driven program using switch-case to input a positive integer and perform one of the following operations:
 1. Find the sum and product of the first and last digits of the number.
 2. Count how many digits in the number are lesser than, equal to, and greater than a given digit k.
- **Input:** A positive integer (num), menu choice (choice), and a single digit (k for Option 2).
- **Output:** Calculated results based on the chosen operation.

2. Algorithm

1. Start
2. Input a positive integer num.
3. Display menu options (1 and 2).
4. Read user choice.

5. Execute operations based on choice using switch-case:

- **Case 1:**

- 1. Extract $\text{last_digit} = \text{num} \% 10$.**
- 2. Set $\text{temp} = \text{num}$. While $\text{temp} \geq 10$, update $\text{temp} = \text{temp} / 10$. Set $\text{first_digit} = \text{temp}$.**
- 3. Calculate $\text{sum} = \text{first_digit} + \text{last_digit}$.**
- 4. Calculate $\text{product} = \text{first_digit} * \text{last_digit}$.**
- 5. Display sum and product.**

- **Case 2:**

- 1. Input a target digit k .**
- 2. Initialize $\text{count_lesser} = 0$, $\text{count_equal} = 0$, $\text{count_greater} = 0$.**
- 3. Set $\text{temp} = \text{num}$.**
- 4. While $\text{temp} > 0$:**
 - Extract $\text{digit} = \text{temp} \% 10$.**
 - If $\text{digit} < k$, increment count_lesser .**
 - Else if $\text{digit} == k$, increment count_equal .**
 - Else, increment count_greater .**
 - Update $\text{temp} = \text{temp} / 10$.**

5. Display count_lesser, count_equal, and count_greater.

- **Default: Print "Invalid Choice!".**

6. End

3. Test Case Table

Test Case	Input Number	Choice	Extra Input	Expected Output	Actual Output	Status
1	12345	1	N/A	Sum = 6, Product = 5	Sum = 6, Product = 5	Pass
2	12345	2	k = 3	Count_lesser = 2, Count_equal = 1, Count_greater = 2	Count_lesser = 2, Count_equal = 1, Count_greater = 2	Pass

Test Case	Input Number	Choice	Extra Input	Expected Output	Actual Output	Status
3	9	1	N/A	Sum = 18, Product = 81	Sum = 18, Product = 81	Pass
4	8080	2	k = 5	Count_lesser = 2, Count_equal = 0, Count_greater = 2	Count_lesser = 2, Count_equal = 0, Count_greater = 2	Pass

4. C Source Code

```
#include <stdio.h>

int main(void) {
    int choice, num, temp, digit, k;
    int first_digit, last_digit, sum, product;
    int count_lesser = 0, count_equal = 0, count_greater = 0;

    printf("Enter a positive integer: ");
    if (scanf("%d", &num) != 1 || num <= 0) {
```

```

    printf("Invalid input! Please enter a positive integer.\n");
    return 1;
}

printf("\n--- MENU ---\n");
printf("1. Sum and Product of the first and last digits\n");
printf("2. Count digits lesser than, equal to, and greater than 'k'\n");
printf("Enter your choice (1 or 2): ");
scanf("%d", &choice);

switch (choice) {
    case 1:
        last_digit = num % 10;
        temp = num;
        while (temp >= 10) {
            temp /= 10;
        }
        first_digit = temp;

        sum = first_digit + last_digit;
        product = first_digit * last_digit;

        printf("\nNumber: %d, Sum = %d, Product = %d\n", num, sum,
product);
        break;

    case 2:
        printf("Enter digit 'k' (0-9): ");
        if (scanf("%d", &k) != 1 || k < 0 || k > 9) {
            printf("Invalid input for k! Enter a single digit between 0 and
9.\n");
            return 1;
        }

        temp = num;
        while (temp > 0) {
            digit = temp % 10;
            if (digit < k) {
                count_lesser++;
            } else if (digit == k) {

```

```

        count_equal++;
    } else {
        count_greater++;
    }
    temp /= 10;
}

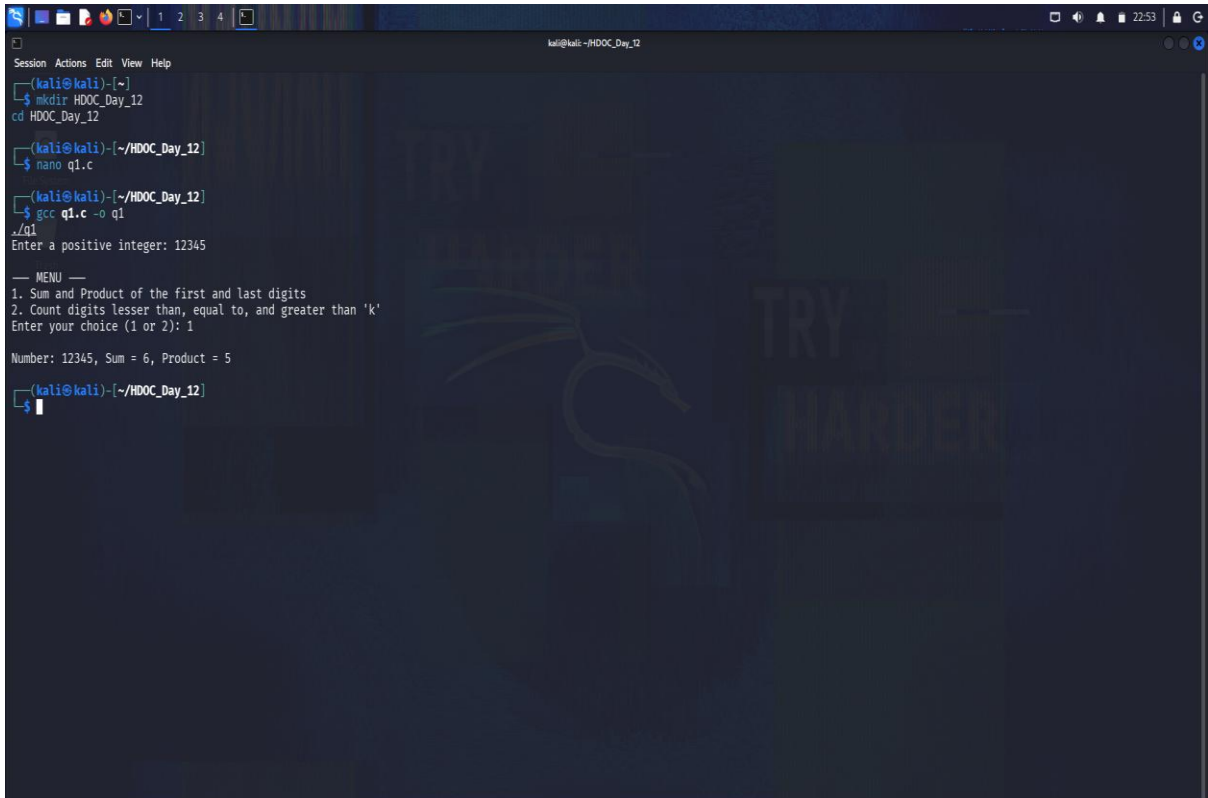
printf("\nNumber: %d, k=%d, Count_lesser=%d,
Count_equal=%d, Count_greater=%d\n",
    num, k, count_lesser, count_equal, count_greater);
break;

default:
    printf("\nInvalid Choice!\n");
    return 1;
}

return 0;
}

```

5. Compilation, Execution & Screenshots



```

kali@kali: ~/HDOC_Day_12
Session Actions Edit View Help
(kali@kali)~$ mkdir HDOC_Day_12
cd HDOC_Day_12
(kali@kali)~/HDOC_Day_12$ nano q1.c
(kali@kali)~/HDOC_Day_12$ gcc q1.c -o q1
./q1
Enter a positive integer: 12345

— MENU —
1. Sum and Product of the first and last digits
2. Count digits lesser than, equal to, and greater than 'k'
Enter your choice (1 or 2): 1

Number: 12345, Sum = 6, Product = 5
(kali@kali)~/HDOC_Day_12$

```